

TPC-383: Local and Pooled Normalization in a Finite $c = 1$ Panel

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Abstract

We test whether the finite origin stability observed in a normalized prime-shell operator is caused by row-wise diagonal whitening. A fresh, response-blind panel with $N = 512$, three origins, three shell anchors, and four fixed laws is evaluated using both local diagonal normalization and a common pooled scalar geometry. The all-plus high- Q origin spread remains below one percent under both choices, while its pooled mean shifts upward by 3.6457251256851203%. The alternating-index control remains unstable, with pooled high- Q spread 10.104585338571119%. This is a finite normalization audit and does not identify an arithmetic source law or prove a twin-prime statement.

1 Question

TPC-382 found a small all-plus magnitude spread across two protocol-matched origin families, but used row-wise normalization. We therefore ask whether the origin-stability shape survives a common scalar normalization, while tracking any absolute calibration shift. The experiment is deliberately finite and response-blind; all four laws are retained as controls.

2 Protocol

The candidate grid is $a_j = 1600001 + 401j$, with indices 0, 20, 40 fixed before any response or metric is read. Thus the origins are (1600001, 1608021, 1616041). Each window has 512 points in four contiguous 128-point blocks. We use the inherited block-distance-one mask, exponent one, beta two, height 66, and $Q \in \{512, 2048, 8192\}$. The laws are all-plus, alternating-index, mod-4 character, and half-split.

Let $M_{o,Q,\ell}$ be the raw law matrix and $g_{o,Q}(i)$ its common square-energy geometry. The two matrices are

$$A_{ij}^{\text{loc}} = M_{ij} / \sqrt{g(i)g(j)}, \quad A_{ij}^{\text{pool}} = M_{ij} / G_Q,$$

where G_Q is the mean of all coordinate geometries over the three fixed origins. The same band mask is applied in both cases. For each law, Q , and normalization, the reported origin spread is

$$\Delta = (\max_o s_o - \min_o s_o) / (\frac{1}{3} \sum_o s_o),$$

with the cap $\Delta \leq 0.01$ fixed in advance.

All current endpoints avoid the prior coordinate panels. The exact $q=8$ anchor [1600001, 1600014] has shell [11, 13]; rational arithmetic verifies positive geometry and symmetry for every law.

3 Results

The local and pooled all-plus high- Q spreads are respectively $1.1394111498671383 \times 10^{-5}$ and $4.6321361430822112 \times 10^{-5}$, both far below the one-percent cap. The pooled scalar is therefore not merely a relabelling: the all-plus high- Q mean changes by

$$\frac{m_{\text{pool}} - m_{\text{loc}}}{m_{\text{loc}}} = 0.036457251256851203.$$

The complete stability census is:

law	local stable cells	pooled stable cells
all-plus	3/3	3/3
alternating-index	0/3	0/3
mod-4 character	3/3	3/3
half-split	3/3	3/3

Thus both normalizations give 9/12 stable cells, but this agreement is law-dependent. At high Q , the pooled alternating-index spread is 0.10104585338571119, the largest in the panel. The result supports a limited shape-versus-calibration separation: all-plus origin stability transfers, while absolute magnitude depends on normalization.

4 Verification and boundary

The producer and a direct reverse-shell independent checker recompute all 72 rows in normal and optimized Python modes. A 25-field adversarial suite rejects altered protocol, row, phase, firewall, and clue fields. The local Bridge-B checks canonical certificate identity, exact anchor data, and byte-identical normal/optimized outputs. The official Session evaluator files are absent, so this is not an official Route-A or Route-B verdict.

claim	status
predeclared panel, mask, and normalization family	PROVED FINITE
72-row local/pooled replay	NUMERICALLY CERTIFIED / FINITE
all-plus high- Q transfer	NUMERICALLY CERTIFIED / FINITE
normalization magnitude shift	NUMERICALLY CERTIFIED / FINITE
source validity and growing uniformity	OPEN
Route-A / Route-B gates	OPEN
twin-prime conclusion	NONE

We record `ARITHMETIC_ADVANCE=NO` and `FIXED_POWER_CREDIT=0`. The next finite question is the bandwidth-normalization phase diagram, which varies the block band while keeping the two normalization rules explicit.